

Lab 09 - Nmap Scanning Techniques



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Lattes

ORCID

SciProfiles

Scopus

ResearcherID

substationworm

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Problem Overview

Containerized Hosts (Generic Corporate Systems)

- server1: 172.30.0.11, 02:aa:bb:cc:dd:11.
- server2: 172.30.0.12, 02:aa:bb:cc:dd:12.
- server3: 172.30.10.10, 02:aa:bb:cc:dd:13.
- attacker: 172.30.0.5 (eth0) and 172.30.10.5 (eth1), 02:aa:bb:cc:dd:01.

Networks

- corp-net: 172.30.0.0/24 (single /24 broadcast domain).
- corp-subnet: 172.30.10.0/26 (subnet).

Tasks

- **1** Issue three ICMP echo requests (ping) from the attacker workstation to server1 while capturing ICMP traffic on server1 using tcpdump. Analyze and describe the communication pattern.
- **2** Conduct a host discovery scan (no port scan, using -sn) on the corp-net from the attacker workstation while monitoring ARP traffic on server1 using tcpdump. Analyze and describe the ARP-based communication.
- **3** Perform a TCP connect scan (-sT) from the attacker workstation against port 22 on server1, and monitor the resulting TCP traffic on server1 using tcpdump.
- **4** Repeat Task 3, but use a SYN stealth scan (-sS) instead of a TCP connect scan.
- **5** Repeat Task 3 using the following TCP flag-based scans: Null scan (-sN), FIN scan (-sF), and Xmas scan (-sX).
- **6** Execute a UDP scan (-sU) from the attacker workstation targeting server2 on ports 53 and 161 while monitoring UDP and ICMP traffic on server2 using tcpdump.
- **7** Use nmap to perform service and version detection (-sV) against ports 22 and 80 on server1.
- **8** Use nmap to conduct a cross-subnet scan (-sS) targeting port 9999 on server3.

Tools

- These are the tools available on the server1, server2, server3, and attacker hosts for completing OTLab 09: ifconfig, nmap, ping, and tcpdump.

Nomenclature

- ARP: Address resolution protocol.
- ICMP: Internet control message protocol.
- IP: Internet protocol.
- MAC: Media access control.
- TCP: Transmission control protocol.
- UDP: User datagram protocol.